filters: *the cutoff frequency of the filter is directly proportional to the clock frequency used to drive the switches*. This makes the switched capacitor filter ideal for data acquisition systems that operate with more than one sampling rate. These are easy-to-use devices; pay ten bucks and have the performance of an eight pole filter inside a single 8 pin IC.

Now for the important part: the characteristics of the three classic filter types. The first performance parameter we want to explore is **cutoff frequency sharpness**. A low-pass filter is designed to block all frequencies above the cutoff frequency (the **stopband**), while passing all frequencies below (the **passband**). Figure 3-11 shows the frequency response of these three filters on a logarithmic (dB) scale. These graphs are shown for filters with a one hertz cutoff frequency, but they can be directly scaled to whatever cutoff frequency you need to use. How do these filters rate? The Chebyshev is clearly the best, the Butterworth is worse, and the Bessel is absolutely ghastly! As you probably surmised, this is what the Chebyshev is designed to do, **roll-off** (drop in amplitude) as rapidly as possible.

Unfortunately, even an 8 pole Chebyshev isn't as good as you would like for an antialias filter. For example, imagine a 12 bit system sampling at 10,000 samples per second. The sampling theorem dictates that any frequency above 5 kHz will be aliased, something you want to avoid. With a little guess work, you decide that all frequencies above 5 kHz must be reduced in amplitude by a factor of 100, insuring that any aliased frequencies will have an amplitude of less than one percent. Looking at Fig. 3-11c, you find that an 8 pole Chebyshev filter, with a cutoff frequency of 1 hertz, doesn't reach an attenuation (signal reduction) of 100 until about 1.35 hertz. Scaling this to the example, the filter's cutoff frequency must be set to 3.7 kHz so that everything above 5 kHz will have the required attenuation. This results in the frequency band between 3.7 kHz and 5 kHz being wasted on the inadequate roll-off of the analog filter.

A subtle point: the attenuation factor of 100 in this example is probably sufficient even though there are 4096 steps in 12 bits. From Fig. 3-4, 5100 hertz will alias to 4900 hertz, 6000 hertz will alias to 4000 hertz, etc. You don't care what the amplitudes of the signals between 5000 and 6300 hertz are, because they alias into the unusable region between 3700 hertz and 5000 hertz. In order for a frequency to alias into the filter's passband (0 to 3.7 kHz), it must be greater than 6300 hertz, or 1.7 times the filter's cutoff frequency of 3700 hertz. As shown in Fig. 3-11c, the attenuation provided by an 8 pole Chebyshev filter at 1.7 times the cutoff frequency is about 1300, much more adequate than the 100 we started the analysis with. The moral to this story: *In most systems, the frequency band between about 0.4 and 0.5 of the sampling frequency is an unusable wasteland of filter roll-off and aliased signals*. This is a direct result of the limitations of analog filters.

The frequency response of the perfect low-pass filter is *flat* across the entire passband. All of the filters look great in this respect in Fig. 3-11, but only because the vertical axis is displayed on a *logarithmic* scale. Another story is told when the graphs are converted to a *linear* vertical scale, as is shown